

Exercise 2:

```
beam_length = 12; % meters
g=@shear_force;
n = 24; % step size is 0.5m, implies n = 24
syms f(x);
f(x) = 5 + 0.25*x^2;
M_analytical = double(int(f,x,0,beam_length));
M_trapezoidal = trapezoidal_rule(g,0,beam_length,n);
M_simpson = simpson(g,0,beam_length,n);
fprintf("Bending moment M analytical= %4.3f \nBending moment M trapezoidal= %4.3f \n" + ...
        "Bending moment M simpson %4.3f \n", ...
        M_analytical,M_trapezoidal,M_simpson)
```

```
Bending moment M analytical= 204.000
Bending moment M trapezoidal= 204.125
Bending moment M simpson 202.313
```

```
function [V] = shear_force(x)
V = 5+ 0.25*x.^2;
end

function [I] = trapezoidal_rule(f,a,b,n)
I=0; % initial value of integral
h = (b-a)/n; % step size
for i=0:n
    x = a + i*h; % going from a to b with 0.5 step size
    if (i==0 || i==n)
        I = I + f(x);
    else
        I = I + 2*f(x);
    end
end
I = I*h/2;
end

function [I] = simpson(f,a,b,n)
if (mod(n,2)==0)
    h=(b-a)/n;
    I=0;
    for i=0:n
        x = a + i*h;
        if(i==0 || i==n)
            I = I + f(x);
        elseif (mod(i,2)==1 && i~=1)
            I = I + 4*f(x);
        else
            I = I + 2*f(x);
        end
    end
    I = I*h/3;
else
```

```
    disp("n must be even")  
end  
end
```